

BATPWN 2020 Crypto: M-ick Maybe

deuterium — 2020-06-08

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/batpwn/crypto/M-ick'Maybe/2020-06-08-BATPWN-Mick-Maybe.html>

Abstract

CTF writeup: Proofreading is cryptanalysis in BATPWN 2020's M-ick Maybe: solve a substitution cipher, then diff it against Moby Dick for stowaway characters.

M-ick Maybe

Challenge

128 Solves



M-ick Maybe

100

Having breakfast and reading moby dick

Author: RevSupport

 ciph3r.txt

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T khbtpdiv exiixlzu rbtw, fsu uzrpzsutsj tswx woz kfm-mxxn fppxrwzu woz jmtsstsj
fifsuixmu gzmz cizfrfswiv. T pozmtrozu sx nfitpz wxlfmur otn, woxbjo oz ofu kzzs
rdvifmdtsj ltwo nz sxw f itwwiz ts woz nfwwzm xe nv kzueziixl.

Oxlzgm, f jxxu ifbjo tr f ntjowv jxxu wotsj, fsu mfwozm wxx rpfmpz f jxxu wotsj; wwoz
nxmz'r woz ctwv. Rx, te fsv xsz nfs, ts otr xls cmxczm czmrxs, feexmu rwbee exm f jxxu
yxdz wx fsvkxuv, izw otn sxw kz kfpdlfmu, kbw izw otn pozzmebiiv fiixl otrnzie wx
rczsu fsu kz rczsw ts wofw lfv. Fsuz woz nfs wofw ofr fsvwotsj kxbswtebiiv ifbjofkiz
fkxbw otn, kz rbmz wozmz tr nxmz ts wofw nfs wofs vxb czmofcr wotsd exm.

Woz kfm-mxxn lfr sxl ebii xe woz ckxfmuzmr lox ofu kzzs umxcctsjs ts woz stjow
cmzgtxbr, fsu loxn T ofu sxw fr vzw ofu f jxxu ixzd fw. Wozv lzmz szfmiv fii lofiznzs;
potze nfwzr, fsu rzpxsu nfwzr, fsu wotmu nfwzr, fsu rzf pfmczswzmr, fsu rzf pxxczmr,
fsu rzf kifpdrntwor, fsu ofmcxxszzmr, fsu rotc dzzczmr; f kmxls fsu kmflsv pxncfsv,
ltwo kxrdv kzfmur; fs bsroxms, rofjjv rzw, fii lzfmstj nxsdzv yfpdzwr exm nxmstsj
jxlsr.

Vxb pxbiu cmzwwv ciftsiv wzii oxl ixsj zfpo xsz ofu kzzs froxmz. Wotr lvxbsj eziixl'r
ozfiwov pozzd tr itdz f rbs-wxfrwzu czfm ts obz, fsu lxbiu rzzn wx rnzii finxrw fr
nbrdv; oz pfssxw ofgz kzzs womzz ufvr ifsuzu emxn otr Tsutfs gxvfjz. Wofw nfs szqw otn
ixxdr f ezl rofuzr itjowzm; vxb ntjow rfv f wxbpo xe rfwtz lxxu tr ts otn. Ts woz
pxncizqtxs xe f wotmu rwtii itsjzmr f wmxctp wfls, kbw ritjowiv kizfpozu ltwofi; oz
uxbkwizrr ofr wfmmztu loxiz lzzdr froxmz. Kbw lox pxbiu roxl f pozzd itdz Hbzzhbzj?
lotpo, kfmzmu ltwo gfmtxbr swtswr, rzznzu itdz woz Fsuzr' lzwzms rixcz, wx roxl exmwo
ts xsz fmmfv, pxswmfrwtjs pitnfwzr, axsz kv axsz.

"Jmbk, ox!" sxl pmtzu woz {ifsuixmu, eitsjtsj xczs f uxxm, fsu ts lz lzw wx
kmzfdefrw.

Wozv rfv wofw nzs lox ofgz rzzs woz lxmiu, wozmkv kzxpnz hbtwz fw zfrz ts nfsszm,
hbtwz rzie-cxrrzrrzu ts pxncfsv. Sxw filfvr, woxbjo: Izuvfmu, woz jmfz Szl Zsjifsu
wmfgzioizm, fsu Nbsjx Cfmd, woz Rpxwpo xsz; xe fii nzs, wozv cxrrzrrzu woz izfrw
frrbmfspszts woz cfmixm. Kbw czmofcr woz nzmz pmxrtsj xe Rtkzmtf ts f rizujz umfls kv
uxjr fr Izuvfmu utu, xm woz wfdtsj f ixsj rxitwfmv lfid xs fs zncwv rxwnfpo, ts woz
szjmx ozfmw xe Femtpf, lotpo lfr woz rbn xe cxxm Nbsjx'r czmexmnfspzr-wottr dtsu xe
wmfgzi, T rfv, nvf sxw kz woz gzmw kzw nxuz xe fwwftstsj f otjo rxptfi cxitro. Rwtii,
exm woz nxrw cfmw, wofw rxmw xe wotsj tr wx kz ofu fsvlozmz.

Wozrz mzeizpwtxsr ybrw ozmz fmz xppfrtxszu kv woz ptmpbnrwfuspsz wofw fewuzm lz lzmz
fii rzfwzu fw woz wfkiz, fsu3 T lfr cmzcfmstj wx ozfm rxnz jxxu rxwmtzr fkbw lofitsj;
wx nv sx rnfiu rbmcmtrz, szfmiv zgzmw nfs nftswftsju f cmxexbsu rtizspz. Fsu sxw xsiv
wofw, kbw wozv ixzdzu znkfmmfrrzu. Vzr, ozmz lzmz f rzw xe srzf-uxjr, nfsv xe loxn
ltwoxbw woz _ritjowzrw kfroeibszrr ofu nkxfmuzu jmfz lofizr xs woz otjo rzfr-zswtmz
rwmfsjzmr wx wozn-fsu ubziizu wozn uzfu ltwoxbw ltsdtsj; fsu vzw, ozmz wozv rfw fw f
rxptfi xkmzfdefrw wfkiz-fii xe woz rfz pfiitsj, fii xe dtsumzu wfwzr-ixxdtsj mxbsu
fr rozzctroiv fw zfpo xwzm fr woxbjo wozv ofu szgzm kzzs xbw xe rtjow xe rxnz
rozzcexiu fnxsj woz Jmzss Nxbswfts. F pbmtxbr rtjow; wozrz kfroeib kzfmr, wozrz
kwtntu lfmmtxm lofiznzs!

Kbw fr exm Hbzzhbzj-lov, Hbzzhbzj rfw wozmz fnxsj wozn-fvw woz ozfu xe woz wfkiz, wxx,
tw rx pofspzu; fr pxxi fr fs ttpiz. Wx kz rbmz T pfssxw rfv nbpo exm otr kmzzutsj.
Otr jmfzfwzrw funtmzm pxbiu sxw ofgz pxmutfiiv ybrwtetzu otr kmstjtsj otr ofmcxxs tswx
kmzfdefrw ltwo otn, fsu brtsj tw wozmz ltwoxbw pzmznxsv; mzfptsj xgzm woz wfkiz ltwo
tw, _wx woz tnntszsw yzxcfmuv xe nfsv ozfur, fsu jmfccitsj woz kzzewzfdr wxlfmur otn.
Kbw wofw lfr pzmwftsiv gzmw pxxiiv uxsz kv otn, fsu zgzmw xsz dsxlr wofw ts nxrw
czxciz'r uzrwtnfwtxs, wx ux fsvwotsj pxxiiv tr wx ux tw tjzswzziiv.

Lz ltii psxw rczfd xe fii Hbzzhbzj'r czpbitfwtwtzr ozmz; oxl oz zrpzlwz pxezz fsu
oxw mxii, fsu fccitzu otr bsutgtuzu fwwzswtxs wx kzzewzfdr, uxsz mfmz. Zsxbjo, wofw
lozs dkmzfdefrw lfr xgzm oz ltwoumzl itdz woz mzw tswx woz cbkitp mxxn, itjowzu otr
wxnfofld-ctcz, fsu lfr rtwtsj wozmz hbtzwiv utjzrwtsj fsu rnxdtsj ltwo otr
tszcfmfkiz ofw xs, lozs T rfiitzu xbw exm f} rwmxi.

We are provided with cipher, we have to solve it. Sounds simple

On a first look, this seems like a monoalphabetic substitution cipher since word sizes are preserved and punctuations are hinting towards that too.

Without wasting any time further, I tried quipquip, which is an amazing cryptogram solver which spits out solved.txt, the contents of which look like :-

I quickly followed suit, and descending into the bar-room accosted the grinning landlord very pleasantly. I cherished no malice towards him, though he had been skylarking with me not a little in the matter of my bedfellow.

However, a good laugh is a mighty good thing, and rather too scarce a good thing; the more's the pity. So, if any one man, in his own proper person, afford stuff for a good joke to anybody, let him not be backward, but let him cheerfully allow himself to spend and be spent in that way. And the man that has anything bountifully laughable about him, be sure there is more in that man than you perhaps think for.

The bar-room was now full of the passengers who had been dropping in the night previous, and whom I had not as yet had a good look at. They were nearly all whalers; chief mates, and second mates, and third mates, and sea carpenters, and sea coopers, and sea blacksmiths, and harpooners, and ship keepers; a brown and brawny company, with bosky beards; an unshorn, shaggy set, all wearing monkey jackets for morning gowns.

You could pretty plainly tell how long each one had been ashore. This young fellow's healthy cheek is like a sun-toasted pear in hue, and would seem to smell almost as musky; he cannot have been three days landed from his Indian voyage. That man next him looks a few shades lighter; you might say a touch of satin wood is in him. In the complexion of a third still lingers a tropic taw, but slightly bleached withal; he doubtless has tarried whole weeks ashore. But who could show a cheek like Queequeg? which, barred with various tints, seemed like the Andes' western slope, to show forth in one array, contrasting climates, zone by zone.

"Grub, ho!" now cried the landlord, flinging open a door, and in we went to breakfast.

They say that men who have seen the world, thereby become quite at ease in manner, quite self-possessed in company. Not always, though: Ledyard, the great New England traveler, and Mungo Park, the Scotch one; of all men, they possessed the least assurance in the parlor. But perhaps the mere crossing of Siberia in a sledge drawn by dogs as Ledyard did, or the taking a long solitary walk on an empty stomach, in the negro heart of Africa, which was the sum of poor Mungo's performances—this kind of travel, I say, may not be the very best mode of attaining a high social polish. Still, for the most part, that sort of thing is to be had anywhere.

These reflections just here are occasioned by the circumstance that after we were all seated at the table, and I was preparing to hear some good stories about whaling; to my no small surprise, nearly every man maintained a profound silence. And not only that, but they looked embarrassed. Yes, here were a set of sea-dogs, many of whom without the slightest bashfulness had boarded great whales on the high seas—entire strangers to them—and duelled them dead without winking; and yet, here they sat at a social breakfast table—all of the same calling, all of kindred tastes—looking round as sheepishly at each other as though they had never been out of sight of some sheepfold among the Green Mountains. A curious sight; these bashful bears, these timid warrior whalers!

But as for Queequeg—why, Queequeg sat there among them—at the head of the table, too, it so chanced; as cool as an icicle. To be sure I cannot say much for his breeding. His greatest admirer could not have cordially justified his bringing his harpoon into breakfast with him, and using it there without ceremony; reaching over the table with it, to the imminent jeopardy of many heads, and grappling the beefsteaks towards him. But that was certainly very coolly done by him, and every one knows that in most

people's destination, to do anything coolly is to do it igenteelly.

We will cnot speak of all Queequeg's peculiarities here; how he eschewed coffee and hot rolls, and applied his undivided attention to beefsteaks, done rare. Enough, that when kbreakfast was over he withdrew like the rest into the public room, lighted his tomahawk-pipe, and was sitting there quietly digesting and smoking with his inseparable hat on, when I sallied out for a} stroll.

As one may observe,

1. There is no direct flag
2. Some spellings are off, e.g. bquickly, cnot and so on

Okay, so the flag is simply the incorrect character in the words with incorrect spellings

Since my lazy ass would not look up for incorrect spellings, I quickly googled the first few lines of the plaintext And I got the text Moby Dick Chapter 5.

Saving the obtained text in mobydick.txt, I can simply write a script, which spits out non-matching words

```
with open('mobydick.txt','r') as f1:
    m1 = f1.read().split()

with open('solved.txt','r') as f2:
    m2 = f2.read().split()

assert len(m1) == len(m2)

for i in range(len(m1)):
    if m1[i] != m2[i]:
        print(m1[i], m2[i])

# quickly bquickly
# landlord alandlord
# the tthe
# more's more's
# boarders pboarders
# young wyoung
# fellow's fellow's
# HE he
# tints, ntints,
# Andes' Andes'
# "Grub, "Grub,
# ho!" ho!"
# landlord, {landlord,
# traveller, travelhler,
# Mungo's Mungo's
# performances-this performances-thiis
# circumstance circumstadnce
# after aftder
# and and3
# sea-dogs, nsea-dogs,
# slightest _slightest
# boarded mboarded
```

```
# breakfast obreakfast
# timid btimid
# them-at them-ayt
# to _to
# THAT that
# people's people's
# estimation, destimation,
# genteelly. igenteelly.
# not cnot
# Queequeg's Queequeg's
# breakfast kbreakfast
# a a}
```

Apart from single quotes, and case differences, we get the flag `batpwn{hidd3n_moby_dick}`

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